Effective sharpness of a pair-moment relaxation for Sidon sets

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Abstract

We give an effective proof of the asymptotic conjecture stated by Akwei for a specified noncyclic pair-moment relaxation of the Sidon problem. The construction is Akwei's renewal boundary profile and capped Fejér covariance, with a positive-part repair and explicit finite error bounds. For every integer $N \geq 2^{60}$, its optimum lies between $\sqrt{N} + (2\sqrt{2}/3)N^{1/4} - 2572$ and $\sqrt{N} + (2\sqrt{2}/3)N^{1/4} + 1$. The lower proof averages the complete renewal cancellation inequality across the centered endpoint atoms, retaining the sign of the renewal jump. It separates a triangular $O(N^{-1/4})$ error from a global $O(N^{-1/2})$ error. The upper bound transfers Chen's finite-measure energy estimate to the relaxation. No numerical feasibility campaign is used.

1 The exact relaxation and attribution

For $N \geq 2$, let $\widehat{h}_2(N)$ maximize $\kappa = \sum_{j=0}^{N-1} f_j$ over real f and symmetric Γ, subject to

$$\begin{pmatrix} 1 & f^{\mathsf{T}} \\ f & \Gamma \end{pmatrix} \succeq 0, \qquad \Gamma_{jj} = f_j, \quad 0 \leq f_j \leq 1, \quad \Gamma_{ij} \geq 0, \tag{1}$$

$$\sum_{j=0}^{N-1-d} \Gamma_{j,j+d} \leq 1 \qquad (1 \leq d < N). \tag{2}$$

This is exactly Akwei's Definition 7.2 [1]. The feasible set is compact: the diagonal is bounded and PSD gives $|\Gamma_{ij}|^2 \leq f_i f_j \leq 1$. A Sidon indicator gives a feasible pair, but a general feasible pair need not be realized by a set or a probability distribution on sets.

Theorem 1. *For every integer $N \geq 2^{60}$,*

$$\sqrt{N} + \frac{2\sqrt{2}}{3}N^{1/4} - 2572 \leq \widehat{h}_2(N) < \sqrt{N} + \frac{2\sqrt{2}}{3}N^{1/4} + 1.$$

The lower bound has the explicit construction given below.

The conjecture, coefficient, renewal profile, centered atoms, covariance ansatz and cap are due to Akwei [1, Section 7]. His preceding Theorem 7.6 is computational at four values of N, and Conjecture 7.7 states the all-N bounded-remainder assertion. The contribution here is the uniform finite-interval analysis, including transition lags and constant objective loss. The upper companion is an attributed consequence of Chen's positive-measure bound [2] and the PSD trace argument already used by Akwei. It does not improve Chen's physical Sidon bound. Hou–Zhao [3] provide the original finite smoothing framework. Constants and onset in this note are deliberately conservative.

2 Renewal bounds and complete cancellation

Let $U = \mathbf{1}_{[0,1)}$ and write $\delta_0 + \sum_{m\geq 1} U^{*m} = \delta_0 + r(t)dt$. The renewal series converges locally by $U^{*m}(t) \leq t^{m-1}/(m-1)!$. Right values are used at jumps. The renewal equation gives

$$r(t) = e^t \ (0 \leq t < 1), \qquad r(t) = e^t - te^{t-1} \ (1 \leq t < 2), \qquad r(t) = \int_{t-1}^{t} r(s)ds \ (t \geq 1).$$

Define $b = (r/2 - 1)\mathbf{1}_{[0,\infty)}$ and $\sigma = (r/2)\mathbf{1}_{[0,\infty)}$.

Lemma 2. *The following bounds hold:*

$$1 \leq r \leq e, \quad \|b\|_\infty \leq \tfrac{1}{2}, \quad \|b\|_1 \leq 4, \quad \mathrm{TV}(b) \leq 10, \quad \int b = -\tfrac{1}{6},$$
$$|b(t)| \leq \tfrac{4}{3}e^{-t/4}, \qquad \int_v^\infty |b(t)|dt \leq 6e^{-v/4}.$$

Writing $A_b(s) = \int_0^\infty b(t)b(t+s)dt$, one has, for $s > 0$,

$$\mathcal{B}(s) := \frac{1}{6} - \int_0^s b + \frac{1}{2}b(s) + A_b(s) \leq \frac{s}{2}. \tag{3}$$

For each real U, the function

$$C_U(s) = U - s - \frac{2}{3} - 2\int_0^s b + 2A_b(s) \tag{4}$$

is 12-Lipschitz on $s > 0$, and $C_U(s) + \sigma(s) \leq U$.

Proof. Here are explicit reconstructions of the renewal estimates and Akwei's cancellation inequality. For $f(z) = r(m - z)$, the Volterra equation on the next unit block gives

$$r(m+y) = \int_0^1 q_y(z)f(z)dz, \quad q_y(z) = \mathbf{1}_{y+z\leq 1} + \int_{\max(0,y+z-1)}^{y} e^s ds.$$

The kernel has mass one and is at least $1/2$ on $0 \leq z \leq 1/2$. Thus the closed block ranges stay in $[1, e]$, are nested, and their oscillations contract by $3/4$. The invariant $\int_{t-1}^t (s - t + 1)r(s)ds = 1$ identifies the limit as 2. Hence $|r(t) - 2| \leq (e-1)(3/4)^{\lfloor t \rfloor}$. This proves the stated norm and tail bounds, using $e < 3$ and $\log(4/3) > 1/4$. The jumps of b at 0 and 1 contribute 1 to its variation. The regular variation on $(0, 1)$ is less than 1, and $b' = b(t) - b(t-1)$ on $t > 1$ contributes at most $2\|b\|_1 \leq 8$.

Put $g = 2b$, $G(t) = \int_0^t g$, and $A_g(s) = \int_0^\infty g(t)g(t+s)dt$. The Laplace transform

$$\widehat{r}(z) = \frac{1 - e^{-z}}{z - 1 + e^{-z}} = \frac{2}{z} - \frac{1}{3} + O(z)$$

and exponential integrability give $\int g = -1/3$. Splitting at the moving jump before differentiation gives, for $0 < s < 1$,

$$A_g'(s) = A_g(s) - A_g(1-s) + e^{1-s} + 4s - 4.$$

Subtracting at $1 - s$ and using antisymmetry gives $A_g(s) - A_g(1-s) = e^s - e^{1-s} - 4s + 2$. Thus $A_g(s) = e^s - 2s + C$. On $s > 1$, $A_g' = A_g(s) - A_g(s-1)$. Decay forces both its moving-integral invariants to vanish, in particular $\int_0^1 sA_g(s)ds = 0$. Consequently $C = -2/3$ and $A_g(0) = 1/3$.

The elementary renewal stopping-time identity gives $G(t) \geq -1$: if T is the first sum of independent uniform increments exceeding t, then $ET < \infty$, $E\sum_{j\leq T} U_j = ET/2 > t$, and $\int_0^t r = ET - 1$. Set $h(s) = 2s + 2 + 2G(s) - g(s) - A_g(s)$. The formulas above yield $h = 8/3$ on $0 < s < 1$. For $s \geq 1$, $h = 2s + 2 + G(s) + G(s-1) - A_g(s)$, with $A_g(s) \leq 1/3$. For $s \geq 3/2$ this proves $h \geq 8/3$. For $1 \leq s \leq 3/2$, $G(s) \geq 3(e-3)/2$, $G(s-1) \geq \sqrt{e} - 2$, so $h \geq 3e/2 + \sqrt{e} - 17/6 > 8/3$. These are exactly (3), including the right value at 1. Finally BV translation gives $\mathrm{Lip}(A_b) \leq 5$, so (4) is 12-Lipschitz and $C_U + \sigma = U - s + 2\mathcal{B} \leq U$. □

The purpose of retaining the whole last inequality is that σ jumps at 1. The proof below translates C_U, which is Lipschitz, rather than Taylor expanding the discontinuous factor.

3 The finite construction and all-distance estimates

Write $x = N^{1/4}$ and fix

$$\ell = \lceil x^2 \rceil + 514, \quad \delta = 1/\ell, \quad H = \lfloor \sqrt{2}x^3 \rfloor, \quad \beta = 1 - 1026/x, \quad c = (\ell - 1)/2. \tag{5}$$

For $x \geq 32768$, $N \geq 8\ell$, $x^3 \leq H \leq \sqrt{2}x^3$, $\ell \leq 5x^2/4$, and $1/2 \leq \beta \leq 1$. Define, for $0 \leq j < N$,

$$\begin{aligned} \psi_j &= \delta[\sigma((j-c)/H) + \sigma((N-1-j-c)/H) - 1]_+, \\ \alpha_j &= \frac{\beta\delta H}{2\ell}\mathbf{1}_{\{j<\ell \text{ or } j\geq N-\ell\}}, \qquad f_j = \psi_j + \alpha_j. \end{aligned} \tag{6}$$

Since $\psi \leq 2\delta$, $f \leq 2/x^2 + 1/x < 1/2$. Put

$$\widetilde{\psi}_j = \min\left\{\psi_j, \sqrt{\frac{\delta(f_j - f_j^2)}{1-\delta}}\right\}, \quad \epsilon_j = f_j - f_j^2 - \frac{1-\delta}{\delta}\widetilde{\psi}_j^2.$$

Let $D_{jj} = \widetilde{\psi}_j/\sqrt{\delta}$, $(\Phi_\ell)_{ij} = (1 - |i-j|/\ell)_+$, and set

$$\Gamma = ff^{\mathsf{T}} + D(I - \delta\Phi_\ell)D + \mathrm{diag}(\epsilon). \tag{7}$$

The Fejér symbol gives $0 \preceq \Phi_\ell \preceq \ell I$. Thus (7) satisfies (1): its covariance is PSD, its diagonal is f, and off-diagonal entries are nonnegative because $0 \leq \widetilde{\psi} \leq f$ and $0 \leq (\Phi_\ell)_{ij} \leq 1$.

All sequences below have zero extension. Write CC, AB, AA for the lag sums of $\psi\psi$, $\alpha\psi + \psi\alpha$, and $\alpha\alpha$, and CT for the lag sum of $\widetilde{\psi}\widetilde{\psi}$. Then, with $w = (1 - d/\ell)_+$ and $S = \delta^2 N$,

$$P_N(d) = CC(d) + AB(d) + AA(d) - wCT(d). \tag{8}$$

Lemma 3. *For $1 \leq d \leq \lfloor N/2 \rfloor$,*

$$P_N(d) \leq S + w\left[\frac{\beta^2\delta^2H^2}{2\ell} - S + \frac{512}{x}\right] + \frac{256}{x^2} + 64e^{-x/24}. \tag{9}$$

Every remaining lag satisfies $P_N(d) < 1$.

Proof. For integrable BV F, lattice quadrature on any shifted mesh of width a has error at most $a\,\mathrm{TV}(F)$. This follows by summing cellwise variation; applying the estimate to $|F|$ proves absolute convergence. In particular

$$D_1 := \sum |\psi_j - \delta| \leq \delta(\ell + 8H + 20) \leq 10\delta H. \tag{10}$$

Indeed the shifted steps miss at most ℓ sites, each sampled b costs at most $4H+10$, and positive part contracts distance from 1. Expanding one factor at a time gives

$$|CC(d) - \delta^2(N-d)| \le 30\delta^2 H, \qquad AB(d) \le 4\delta^2 H. \tag{11}$$

Monotonicity of $t(1-t)$ on $[0,1/2]$ proves $\widetilde{\psi} \ge \min(\psi, \delta)$. Hence $\sum(\psi - \widetilde{\psi}) \le D_1$, $CC - CT \le 40\delta^2 H$, and for $d < \ell$,

$$S - CT(d) \le 71\delta^2 H. \tag{12}$$

Exact interval counting, with $a_0 = \beta\delta H/(2\ell)$, gives

$$AA(d) = a_0^2\{2(\ell-d)_+ + (\ell - |d-(N-\ell)|)_+\}. \tag{13}$$

It remains to improve (11) away from very short lags. Set $U = (N-\ell)/H$ and $\chi_U(z) = [1 + b(z) + b(U-z)]\mathbf{1}_{[0,U]}(z)$. Then $0 \le \chi_U \le 2$, $\mathrm{TV}(\chi_U) \le 26$, so the product $\chi_U(z)\chi_U(z+s)$ has variation at most 104. The difference between ψ_j and $\delta\chi_U((j-c)/H)$ is nonnegative and supported on at most ℓ outer sites; its total mass is at most $(4/3)e^{-U/4}$. Its cost to the complete un-capped pair sum is at most $8e^{-U/4}$.

For $\ell/2 \le d \le N/2$, put $s = d/H$, $V = U - s \ge N/(4H)$. Expanding the finite continuous autocorrelation gives

$$\left| \int \chi_U(z)\chi_U(z+s)dz - C_U(s) \right| \le (30 + 4U)e^{-V/4}. \tag{14}$$

The two linear tails and two same-end product tails cost at most $5\int_V^\infty |b| \le 30e^{-V/4}$. The two cross-end products cost at most $(16/9)Ve^{-V/4} + (16/9)Ve^{-(U+s)/4} \le 4Ue^{-V/4}$. Discrete continuous/continuous quadrature costs at most $104\delta^2$.

Let $h_j = (j-c)/H$, $0 \le j < \ell$. These shifts have mean zero and $|h_j| \le \ell/(2H)$. Exact reflection leaves two inward atom terms, with normalization $\beta\delta^2 H$ times the average $\chi_U(s+h_j)$. Outward atom/χ_U terms vanish because $\lceil c \rceil + \lceil \ell/2 \rceil = \ell$. All $s + h_j > 0$; their remote boundary corrections have magnitude at most $(4/3)e^{-N/(16H)}$. Since $\beta \le 1$ and $\sigma \ge 0$, average the *whole* inequality in Lemma 2 to obtain

$$C_U(s) + \frac{1}{\ell}\sum_j \sigma(s+h_j) \le U + 12\frac{1}{\ell}\sum_j |h_j| \le U + 6\ell/H.$$

This argument retains the renewal jump rather than differentiating it. Together with (14), it yields

$$CC(d) + AB(d) \le S + 109/x^2 + \mathcal{T}_N, \qquad \mathcal{T}_N = 64e^{-x/24}. \tag{15}$$

For completeness, put $z = N/H$, so $x/\sqrt{2} \le z \le x$. The total tail ledger is at most $[\delta^2 H(32+4z) + 8]e^{-z/16} \le 32e^{-x/24}$, leaving slack in $\mathcal{T}_N$.

For $d < \ell/2$, $w \ge 1/2$, and (11) gives $CC + AB \le S + 136w/x$. For $\ell/2 \le d < \ell$, use (15). Equation (12) costs at most $142w/x$. For $d \ge \ell$, the hole vanishes and (15) applies directly. Thus (8) and (13) prove (9), with room between 278 and 512 and between 109 and 256.

For $N/2 < d \le N - 2\ell$, there is no atom/atom term, and $P_N(d) \le 1/2 + 68/x < 1$ by (11). For $N - 2\ell + 1 \le d < N$, the opposite atom triangle is at most $1/2$, the continuous overlap is at most $8\delta^2\ell$, and $AB \le 4\delta^2 H$. Hence $P_N(d) \le 1/2 + 8/x^2 + 8/x < 1$. This is an exhaustive integer partition; the opposite strip peaks at $d = N - \ell$. □

4 Parameter absorption and objective

The explicit selection rule is $B_0 = \lceil 2(512+1) \rceil = 1026$, $D_0 = \lceil 2(256+1) \rceil = 514$. For $x \geq 32768$, $x \geq 2B_0$, $x^2 \geq 4(D_0+1)$, and

$$\frac{1}{2} \leq S \leq 1 - \frac{514}{x^2}, \qquad \frac{H^2}{2\ell} \leq N, \qquad 1-\beta^2 \geq \frac{1026}{x}.$$

For the upper estimate on S, put $q = (\ell - x^2)/x^2 \leq 1/4$; then $1-(1+q)^{-2} \geq q \geq 514/x^2$. The square bracket in (9) is therefore at most $-1/x$. The elementary exponential bound gives

$$64e^{-x/24} \leq \frac{64\,24^5}{x^4} \leq x^{-2}, \qquad 64\,24^5 < 2^{30} \leq x^2.$$

Thus all short and medium budgets are at most $1 - 257/x^2 - w/x < 1$, and Lemma 3 covers the other lags. The construction is feasible for every integer $N \geq 2^{60}$.

Positive part adds mass, while the cap changes only Γ. Using the BV quadrature bound and $\int b = -1/6$ gives

$$\sum_j \psi_j \geq \delta H \left(U - \frac{1}{3} - 2\int_U^\infty |b| \right) - 26\delta.$$

The atoms have total mass $\beta\delta H$ and $HU = N - \ell$. Consequently

$$\kappa \geq \delta N - 1 + \frac{2}{3}\delta H - \frac{1026\delta H}{x} - 26\delta - 12\delta H e^{-U/4}. \tag{16}$$

Here $\delta N \geq x^2 - 515$, $\delta H \geq \sqrt{2}x - 1$, $1026\delta H/x < 2052$, and $26\delta < 1$. Also $U \geq x/2$, and $e^{-U/4} \leq 24\,8^4/x^4 \leq x^{-2}$, so the last loss is at most $24/x < 1$. The losses in (16) total at most 2571. The declared constant 2572 includes one further unit of slack.

5 An attributed uniform upper companion

Let $\phi(t) = 4/3 - 2|t| + (2/3)|t|^3$ on $|t| \leq 1$, zero otherwise. It is the autocorrelation of a normalized ramp. Thus $A_T(i,j) = \phi((i-j)/T)$ is PSD. For any feasible moment pair,

$$f^{\mathsf{T}} A_T f \leq \operatorname{tr}(A_T \Gamma) \leq \frac{4}{3}\kappa + 2\sum_{d \geq 1} \phi(d/T) \leq \frac{4}{3}\kappa + T.$$

The last estimate uses the decreasing nonnegative kernel and its unit integral, and holds for real $T > 0$. Chen's finite-positive-measure comparison [2, Lemma 4.4], applied to $\mu = \sum_j f_j \delta_{j/T}$ on $[0, N/T]$, yields

$$\kappa^2 \leq \left(\frac{N}{T} + \frac{2}{3} + 200e^{-aN/T} \right) \left(T + \frac{4}{3}\kappa \right), \qquad a = \log(4/3), \quad 0 < T \leq N. \tag{17}$$

This uses energy Cauchy–Schwarz with a finite signed reference measure whose potential is exactly one on the closed interval. Its energy is at most $N/T + 2/3 + 200e^{-aN/T}$. It imposes no integrality on κ, and no division by κ is required.

For completeness, a simple large-onset root comparison suffices here. Set $T = \sqrt{2}x^3$, $g = 2\sqrt{2}/3$, and $\eta = 200e^{-ax/\sqrt{2}}$. Equation (17) says

$$P_\eta(\kappa) \leq 0, \quad P_\eta(z) = z^2 - (gx + 8/9 + 4\eta/3)z - x^4 - gx^3 - \sqrt{2}\eta x^3.$$

At $y = x^2 + gx + 1$, direct expansion gives $P_0(y) = (10/9)x^2 + (g/9)x + 1/9$. Since $a > 1/4$ and $\sqrt{2} < 3/2$,

$$\eta \le 200e^{-x/6} \le \frac{200 \cdot 24 \cdot 6^4}{x^4} \le x^{-2} \quad (x \ge 32768).$$

The error at y is at most $4 + 2x < (10/9)x^2$. Hence $P_\eta(y) > 0$. The quadratic has negative constant term and one positive root, so $\kappa < y$, proving the upper half of Theorem 1. Chen's sharper rational absorption gives this upper companion already for $N \ge 4{,}600{,}000$; its all-real-mass replay is retained in the evidence packet. The lower onset 2^{60} dominates it.

6 Scope, reproducibility and review status

The theorem concerns only (1)–(2). It does not exclude stronger pair models containing further valid quadratic or distinct-sum constraints. Covariance PSD is not a realizability theorem [4]. No claim about a Sidon-set lower construction, a coefficient below the smoothing floor, or a resolution of Erdős problem 30 follows.

The accompanying evidence packet contains the frozen statement, three complete proof sections, two fresh independent internal mathematical reviews with source hashes, objection resolutions, and a separate bounded primary-source comparison. No optimizer, FFT, sampled feasibility, renewal table or numerical campaign is used. Existing rational smoothing certificates and public releases are unchanged.

AI assistance and limitations. This draft, its derivation and its internal mathematical reviews were prepared with assistance from OpenAI Codex. Internal AI review is not external human refereeing or proof-kernel formal verification. The literature audit is bounded and does not establish worldwide priority. This is a public attributed preprint. External human verification and peer review remain pending; public availability does not constitute external acceptance.

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